

Sudhakar Vuppalapati

Principal Solutions Architect | AI Orchestration & Cloud | AWS Certified | 18+ Years

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PROFESSIONAL SUMMARY

Results-driven Principal Solutions Architect with 18 years of experience designing and delivering cloud-native, event-driven, and distributed systems at enterprise scale. Deep expertise in AWS architecture, microservices, Kubernetes (EKS), and streaming platforms. Now expanding into Generative AI engineering — building AI-orchestrated workflows with Anthropic Claude, MCP servers, Claude Code, and GitHub Actions on Azure. Proven track record across Automotive, Retail, Telematics, Aviation, and IoT domains. AWS Certified Solutions Architect with a filed US patent.

AI ENGINEERING & EMERGING SKILLS

Actively upskilling in Generative AI engineering and agentic systems. Hands-on experience building AI-orchestrated workflows using Anthropic Claude, Claude Code CLI, and Model Context Protocol (MCP) servers integrated with Spring Boot microservices and automated via Azure-hosted GitHub Actions CI/CD pipelines.

AI Orchestration (Agentic)	Claude Code CLI	MCP Server Integration
LLM API Integration	GitHub Actions CI/CD	Azure DevOps Pipelines
Spring Boot + AI APIs	Prompt Engineering	Event-Driven AI Workflows

- Designed and prototyped AI-orchestrated workflows using **Anthropic Claude** APIs integrated with Spring Boot microservices, enabling intelligent event-driven processing pipelines.
- Hands-on experience with **Claude Code CLI** — an agentic coding tool — for AI-assisted software development, code generation, refactoring, and multi-file project navigation.
- Implemented **Model Context Protocol (MCP)** servers to expose structured tool interfaces to LLMs, enabling Claude to interact with external APIs, databases, and cloud services within multi-agent orchestration flows.
- Built and configured **GitHub Actions CI/CD pipelines on Azure** to automate build, test, and deployment of AI-integrated Spring Boot services — including MCP server deployments.
- Explored multi-agent orchestration patterns: chaining LLM calls, tool-use loops, and human-in-the-loop approval flows using Claude's tool-use API within Spring Boot applications.
- Applied prompt engineering and structured output techniques to integrate LLM reasoning into existing microservice architectures without disrupting established API contracts.

CORE ARCHITECTURE COMPETENCIES

Cloud-Native Architecture	Microservices & REST APIs	Event-Driven Architecture
AWS EKS / ECS Fargate	Apache Kafka / Kinesis / Flink	Infrastructure as Code (CFN)
System Design & Patterns	CI/CD & DevOps	IoT & Streaming Platforms
Java / Spring Boot / Scala	Docker & Kubernetes	Data Migration & Modernisation

PROFESSIONAL EXPERIENCE

Senior Technical Architect

Cognizant Technology Solutions | Feb 2025 – Present

- Architected a secure AWS Transfer Family SFTP-to-S3 ingestion platform with event-driven CSV processing, reliable database synchronisation, and enterprise-grade error handling.
- Automated infrastructure provisioning with AWS CloudFormation (IaC) integrated into Azure DevOps CI/CD pipelines.
- Delivered a reusable, configurable event-processing framework supporting consumer-specific business logic at scale.
- Enforced security and compliance via IAM access controls, audit logging, automated backups, and disaster recovery.

Technical Architect

Delta Airlines | Oct 2023 – Jul 2024

- Designed real-time duplicate booking detection using AWS MSK (Managed Kafka) and AWS Step Functions — significantly reducing duplicate records and improving data reliability.
- Built a secure SFTP-to-S3 ingestion pipeline with event-driven CSV processing and robust exception handling.
- Collaborated with product and business teams to translate customer service goals into scalable technical solutions.

Principal Architect

HERE Technologies | Aug 2021 – Oct 2023

- Led solution architecture for private mapping and ISA compliance for Automotive OEM clients (VINFAST, Toyota).
- Designed C++ client-side applications on QNX platform consuming HERE REST APIs for real-time map data.
- Migrated Toyota MM21 connected car microservices from on-premises Kubernetes to AWS EKS with zero service disruption.
- Integrated Nuance NLP platform APIs to build location and driver-behaviour context solutions.

Technical Manager / Architect

Infosys | Feb 2013 – Apr 2021

- Designed LexisNexis Global Telematics Platform (GTP) on AWS — Kinesis Firehose ingestion, ECS Fargate enrichment, and data-driven insurance premium scoring for GM (US) and Fiat/Chrysler (EU).
- Architected AGCO farm management platform on AWS ECS Fargate using CloudFormation IaC and multi-vendor integrations.
- Built a computer vision analytics platform using Spring Boot, Apache Kafka, OpenCV, and Apache Flink.
- Filed US Patent Application No. 17/513,597 — distributed in-memory RDBMS query engine (Akka/Scala).

Staff System Software Engineer (Product Owner)

IBM India Software Labs | Apr 2011 – Feb 2013

- Product Owner for Asterix IBM POWER virtualisation tool; aligned HMC and hardware development roadmaps.
- Managed end-to-end SDLC using IBM Rational tooling across distributed hardware and software teams.

R&D; Engineer

Nokia India Pvt Ltd | May 2007 – Mar 2011

- Implemented JSR specs for Nokia S60 Symbian Java runtime: MIDP, PIM API, WMA, Mobile Media API, and AMMS.

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate (SAA-C03)
- HiveMQ MQTT Associate Certification
- Certified Scrum Master (CSM) — Scrum Alliance
- Sun Certified Java Programmer (SCJP 1.5) & Mobile Application Developer (SCMAD 1.0)
- Introduction to People Management — IIM Bangalore Executive Programme

EDUCATION

B.Tech, Computer Science & Information Technology

G. PullaReddy Engineering College, Kurnool | Sri Krishnadevaraya University

Diploma, Electronics & Communication Engineering

Sri Venkateswara Govt Polytechnic, Tirupati